

Chapter 7: Distributed Systems Consensus

Distributed consensus ensures that multiple nodes agree on a state value.

The Raft consensus protocol achieves state machine replication via an elected leader.

Raft decomposes consensus into leader election, log replication, and safety guarantees.

Nodes in Raft exist in one of three states: Leader, Follower, or Candidate.

Comparative Analysis of Consensus Protocols

The following table details protocol complexities and guarantees

Protocol	Leader Model	Fault Tolerance	Understandability
Raft	Strong Single Leader	Majority ($N/2 + 1$)	High
Multi-Paxos	Weak/Multi Leader	Majority ($N/2 + 1$)	Low
Zab (ZooKeeper)	Primary-Backup	Majority ($N/2 + 1$)	Moderate
PBFT	View-Change Leader	$(3f + 1)$ nodes	Complex

A Raft cluster of 5 nodes can continue operating properly despite 2 node failures.

Raft Node State Machine Diagram

[Follower State] --(election timeout)--> [Candidate State]

[Candidate State] --(votes from majority)--> [Leader State]

[Candidate State] --(discovers new leader)--> [Follower State]

[Leader State] --(discovers term > currentTerm)--> [Follower State]

Heartbeats: The Leader sends periodic AppendEntries RPCs with no log entries.

If followers receive heartbeats before their election timeout, they remain followers.

Byzantine Fault Tolerance and Network Partitions

Crash fault tolerance assumes nodes fail by stopping, not by lying or sending corrupt packets.

Raft and Paxos handle crash fault tolerance (CFT).

During a network partition, only the partition containing a strict majority can commit entries.

The minority partition cannot achieve quorum and rejects write requests.